

$^{24}\text{Mg}(^{14}\text{N},3\text{p}\gamma)$ 2014Ay01

2014Ay01: $^{24}\text{Mg}(^{14}\text{N},3\text{p})^{35}\text{S}$ fusion-evaporation reaction. A 40-MeV, 5-pnA ^{14}N beam was delivered from the LNL XTU-Tandem accelerator and impinged on 1-mg/cm² thick, 99.7% isotopically enriched ^{24}Mg evaporated on an 8-mg/cm² gold layer. γ rays were detected using 4 π -GASP array composed of 40 Compton-suppressed large volume HPGe detectors arranged in seven rings at different angles with respect to beam axis. Events were collected when at least two germanium detectors fired in coincidence. Measured E_γ , I_γ , $\gamma\gamma$ -coin, $\gamma(\theta)$, $\gamma\gamma(\theta)$ (ADO). Deduced levels, J , π , level lifetimes via the Doppler shift attenuation method (DSAM). Compared with large-scale shell model calculations.

 ^{35}S Levels

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>$T_{1/2}$[#]</u>	<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>$T_{1/2}$[#]</u>
0.0 ^{&}	3/2 ⁺	3816.0 [@] 3	9/2 ⁻	0.28 ps 3	5412.2 3	9/2 ⁺	
1572.5 3	1/2 ⁺	3886.4 4	5/2 ⁺		5877.6 ^{&} 3	11/2 ⁺	
1991.4 [@] 2	7/2 ⁻	4023.2 [@] 3	11/2 ⁻	0.32 ps 3	6299.4 4	11/2 ⁺	
2347.9 3	3/2 ⁻	4822.9 3	9/2 ⁺		6352.2 [@] 5	13/2 ⁻	50 fs 10
2717.1 3	5/2 ⁺	4899.7 4	9/2 ⁺		7179.8 ^{&} 3	15/2 ⁺	3.1 ps 12
3594.7 ^{&} 3	7/2 ⁺	5010.2 3	11/2 ⁻	0.45 ps 8	8023.7 ^{&} 9	17/2 ⁺	0.15 ps 4

[†] From a least-squares fit to γ -ray energies.

[‡] As given in 2014Ay01, which quotes the known J^π assignments for four low-lying levels at 1572, 1991, 2347, and 2717 from 2011Ch48. 2014Ay01 assigns J^π for levels above 3 MeV based on the measured $\gamma\gamma(\theta)$ (ADO) ratios. When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

[#] From DSAM line-shape analyses (2014Ay01).

[@] Seq.(A): $\Delta J=1$ sequence based on 7/2⁻.

[&] Seq.(B): Sequence based on 3/2⁺.

 $\gamma(^{35}\text{S})$

$R_{\text{ADO}}=[I_\gamma(34^\circ)+I_\gamma(146^\circ)]/2I_\gamma(90^\circ)$. Expected values are 0.8 for stretched dipole ($\Delta J=1$) and 1.4 for stretched quadrupole ($\Delta J=2$) or non-stretched dipole ($\Delta J=0$) transitions.

<u>E_γ</u>	<u>I_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[†]</u>	<u>Comments</u>
207.0 5	0.4 2	4023.2	11/2 ⁻	3816.0	9/2 ⁻		
465.3 3	1.2 1	5877.6	11/2 ⁺	5412.2	9/2 ⁺	D	$R_{\text{ADO}}=0.71$ 9.
775.4 4	1.4 2	2347.9	3/2 ⁻	1572.5	1/2 ⁺	D	$R_{\text{ADO}}=0.68$ 14.
827.8 6	0.3 1	7179.8	15/2 ⁺	6352.2	13/2 ⁻		
843.9 8	1.6 2	8023.7	17/2 ⁺	7179.8	15/2 ⁺	D	$R_{\text{ADO}}=0.8$ 3.
867.3 3	1.9 1	5877.6	11/2 ⁺	5010.2	11/2 ⁻	D	$R_{\text{ADO}}=1.36$ 20, consistent with $\Delta J=0$.
877.6 5	0.4 1	3594.7	7/2 ⁺	2717.1	5/2 ⁺		
880.2 4	0.8 1	7179.8	15/2 ⁺	6299.4	11/2 ⁺		
887.0 7	0.4 1	6299.4	11/2 ⁺	5412.2	9/2 ⁺	D	$R_{\text{ADO}}=0.6$ 3.
977.8 3	1.6 1	5877.6	11/2 ⁺	4899.7	9/2 ⁺	D+Q	$R_{\text{ADO}}=1.43$ 16.
986.9 3	4.4 2	5010.2	11/2 ⁻	4023.2	11/2 ⁻	D	$B(E1)(\text{W.u.})=0.00102 +22-16$; $B(M1)(\text{W.u.})=0.036 +8-6$ $R_{\text{ADO}}=1.30$ 13, consistent with $\Delta J=0$.
1006.8 2	0.8 1	4822.9	9/2 ⁺	3816.0	9/2 ⁻	D	$R_{\text{ADO}}=1.37$ 11, consistent with $\Delta J=0$.
1054.5 3	3.0 2	5877.6	11/2 ⁺	4822.9	9/2 ⁺	D	$R_{\text{ADO}}=0.72$ 11.
1228.1 3	1.1 1	4822.9	9/2 ⁺	3594.7	7/2 ⁺	D	$R_{\text{ADO}}=0.78$ 9.
1302.2 2	13.5 8	7179.8	15/2 ⁺	5877.6	11/2 ⁺	E2	$B(E2)(\text{W.u.})=6.6 +42-19$ $R_{\text{ADO}}=1.40$ 5, $A_2=+0.27$ 5, $A_4=-0.07$ 5.
1304.8 12	0.6 2	4899.7	9/2 ⁺	3594.7	7/2 ⁺		
1389.0 4	3.9 4	5412.2	9/2 ⁺	4023.2	11/2 ⁻	D	$R_{\text{ADO}}=0.79$ 11.

Continued on next page (footnotes at end of table)

$^{24}\text{Mg}(^{14}\text{N}, 3\text{p}\gamma)$ **2014Ay01** (continued) $\gamma(^{35}\text{S})$ (continued)

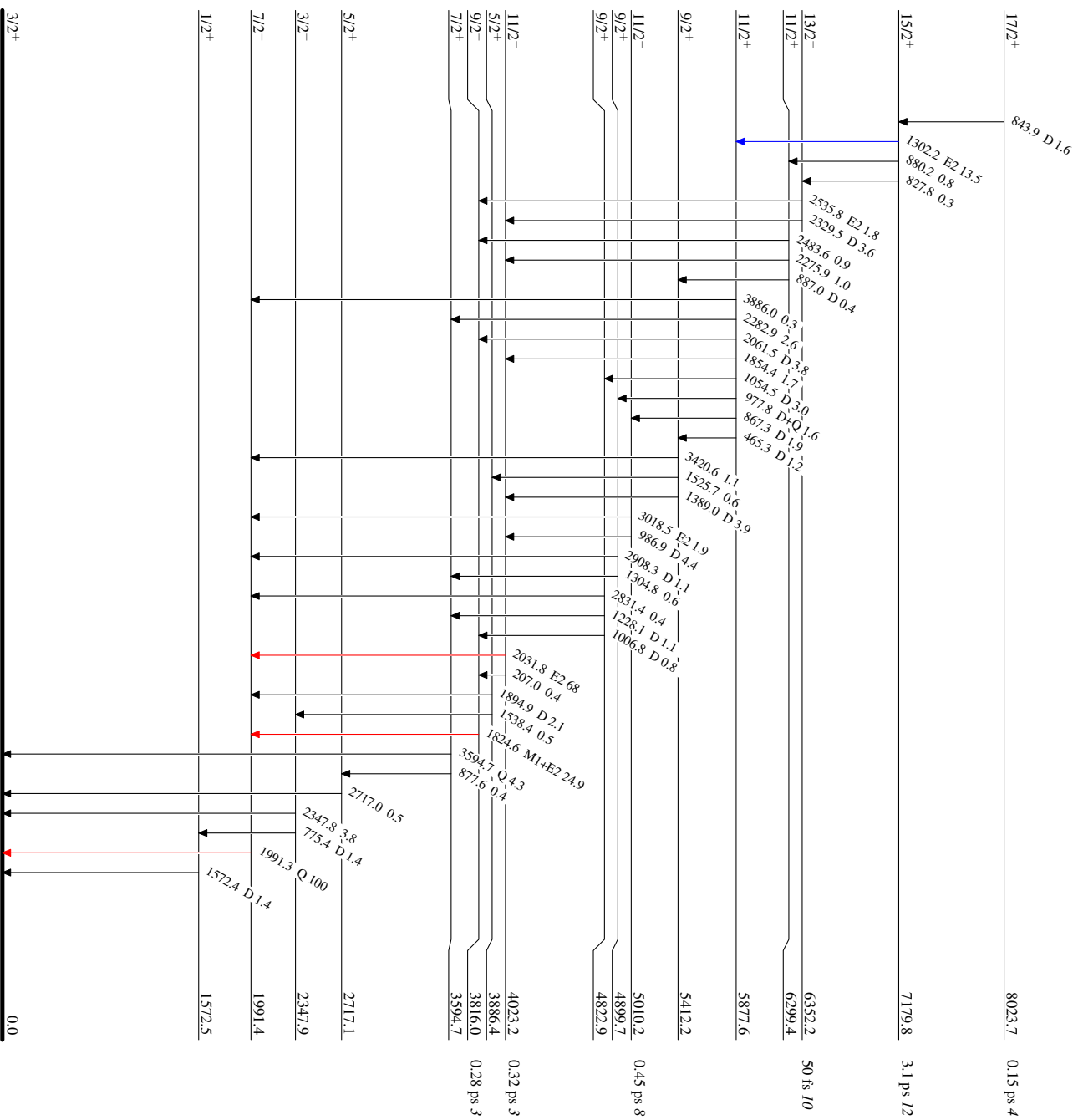
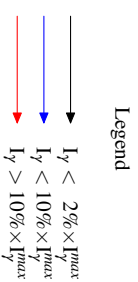
E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [†]	$\delta^\dagger$	Comments
1525.7 8	0.6 1	5412.2	9/2 ⁺	3886.4	5/2 ⁺			
1538.4 8	0.5 2	3886.4	5/2 ⁺	2347.9	3/2 ⁻			
1572.4 3	1.4 4	1572.5	1/2 ⁺	0.0	3/2 ⁺	D		$R_{\text{ADO}}=0.88$ 10.
1824.6 2	24.9 15	3816.0	9/2 ⁻	1991.4	7/2 ⁻	M1+E2	+0.55 9	$B(\text{M1})(\text{W.u.})=0.0099$ +14-12; $B(\text{E2})(\text{W.u.})=3.4$ +10-9 Mult., δ : D+Q and δ from $\gamma(\theta)$ in 2014Ay01 ; E1+M2 ruled out by RUL. $R_{\text{ADO}}=1.58$ 5 indicating $\Delta J=2$ or 0 seems questionable considering $\Delta J=1$ from level scheme. $A_2=+0.48$ 4, $A_4=+0.08$ 6.
1854.4 4	1.7 1	5877.6	11/2 ⁺	4023.2	11/2 ⁻			
1894.9 4	2.1 3	3886.4	5/2 ⁺	1991.4	7/2 ⁻	D		$R_{\text{ADO}}=0.69$ 10.
1991.3 2	100 6	1991.4	7/2 ⁻	0.0	3/2 ⁺	Q		$R_{\text{ADO}}=1.45$ 15.
2031.8 3	68 4	4023.2	11/2 ⁻	1991.4	7/2 ⁻	E2		$B(\text{E2})(\text{W.u.})=7.5$ +8-7 $R_{\text{ADO}}=1.41$ 10, $A_2=+0.29$ 4, $A_4=-0.10$ 6. $R_{\text{ADO}}=0.65$ 9.
2061.5 3	3.8 2	5877.6	11/2 ⁺	3816.0	9/2 ⁻	D		
2275.9 6	1.0 2	6299.4	11/2 ⁺	4023.2	11/2 ⁻			
2282.9 4	2.6 1	5877.6	11/2 ⁺	3594.7	7/2 ⁺			
2329.5 9	3.6 7	6352.2	13/2 ⁻	4023.2	11/2 ⁻	D		$B(\text{E1})(\text{W.u.})=6.7 \times 10^{-4}$ +17-13; $B(\text{M1})(\text{W.u.})=0.023$ +6-5 $R_{\text{ADO}}=0.8$ 3.
2347.8 3	3.8 3	2347.9	3/2 ⁻	0.0	3/2 ⁺			
2483.6 8	0.9 2	6299.4	11/2 ⁺	3816.0	9/2 ⁻			
2535.8 11	1.8 2	6352.2	13/2 ⁻	3816.0	9/2 ⁻	E2		$B(\text{E2})(\text{W.u.})=5.3$ +17-11 $R_{\text{ADO}}=1.5$ 5.
2717.0 4	0.5 3	2717.1	5/2 ⁺	0.0	3/2 ⁺			
2831.4 9	0.4 1	4822.9	9/2 ⁺	1991.4	7/2 ⁻			
2908.3 5	1.1 2	4899.7	9/2 ⁺	1991.4	7/2 ⁻	D		$R_{\text{ADO}}=0.74$ 14.
3018.5 7	1.9 2	5010.2	11/2 ⁻	1991.4	7/2 ⁻	E2		$B(\text{E2})(\text{W.u.})=0.223$ +49-39 $R_{\text{ADO}}=1.35$ 34.
3420.6 9	1.1 1	5412.2	9/2 ⁺	1991.4	7/2 ⁻			
3594.7 7	4.3 4	3594.7	7/2 ⁺	0.0	3/2 ⁺	Q		$R_{\text{ADO}}=1.36$ 16.
3886.0 13	0.3 1	5877.6	11/2 ⁺	1991.4	7/2 ⁻			

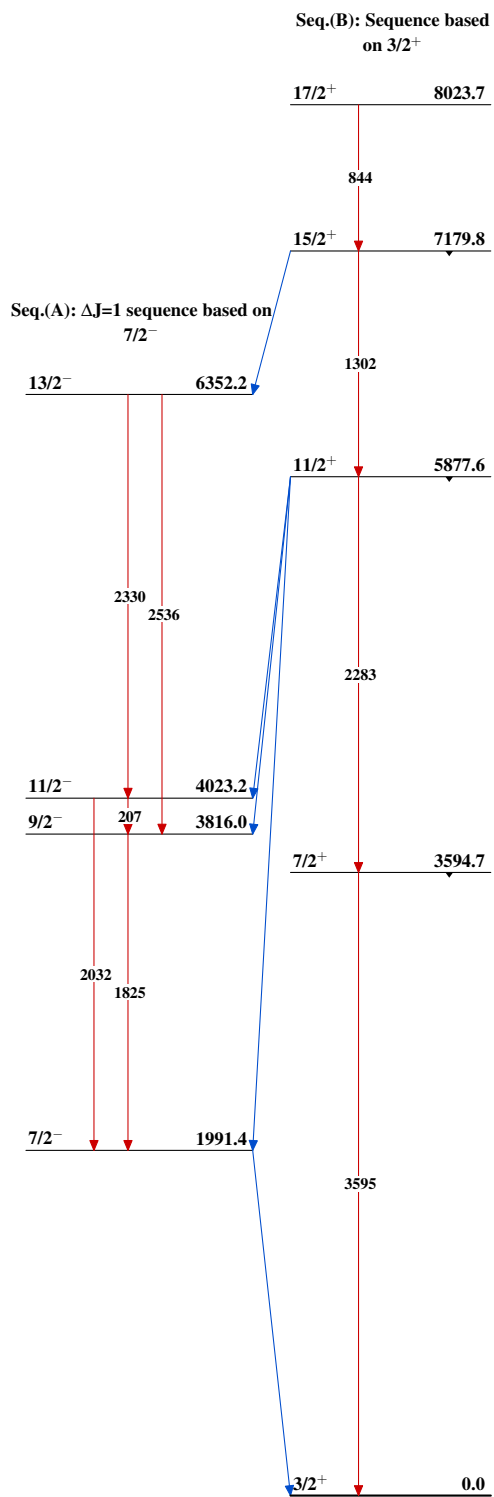
[†] From $\gamma(\theta)$, $\gamma\gamma(\theta)$ (ADO) data, and RUL when $T_{1/2}$ is available in [2014Ay01](#).

²⁴Mg(¹⁴N,3pγ) 2014Ay01

Level Scheme

Intensities: Relative I_γ



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